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Network Topology Optimization for Bearing Rigid Multi-Agent Systems via Deep Reinforcement Learning and Graph Neural Networks

Department of Information Engineering
Master's Degree in Control Systems Engineering

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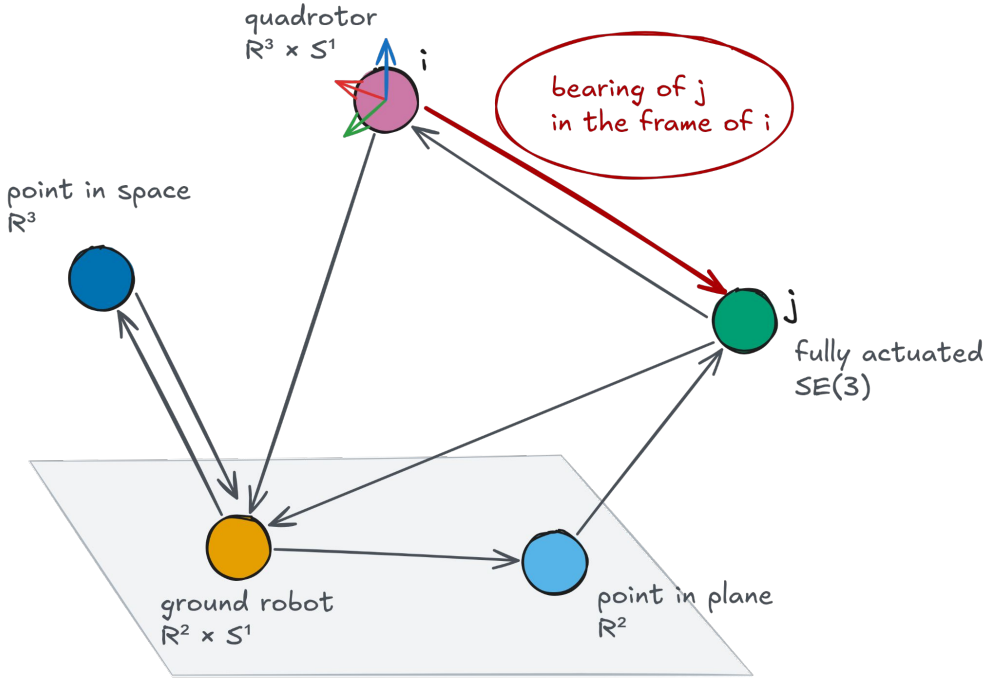
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Academic Year 2025/2026

Outline

1. **Bearing rigidity and motivation**
2. **Rigidity analysis for heterogeneous formations**
   rigidity matrix, edge-count bound, single-edit quantities, estimation error
3. **Topology optimization with reinforcement learning**
   structure of the objectives, sequential formulation for Graph RL
4. **Experiments and results**
   comparison with classical search, trade-off, generalization, cost
5. **Conclusions**

Bearing Rigidity, Heterogeneous Agents



- Rigid (IBR): formation shape is recoverable from bearings

$$(\mathcal{G}, \chi)\ \text{IBR} \iff \operatorname{rk} \mathbf{B}_{\mathcal{G}} = \operatorname{rk} \mathbf{B}_{\mathcal{K}}$$

- Rigidity eigenvalue:

$$\lambda = \mu_{6n-\rho_{\mathcal{K}}+1}(\mathbf{B}^\top \mathbf{B}) = s^2_{\rho_{\mathcal{K}}}, \qquad \|\mathbf{B}\,\delta\chi\| \geq \sqrt{\lambda}\,\|\delta\chi\| \ \ \forall\, \delta\chi \perp \ker \mathbf{B}$$

- Estimation: bearing error amplified by at most $1/\sqrt{\lambda}$
- Convergence rate of bearing-only localization

Extended rigidity matrix

- Rigidity matrix: Jacobian of the bearings

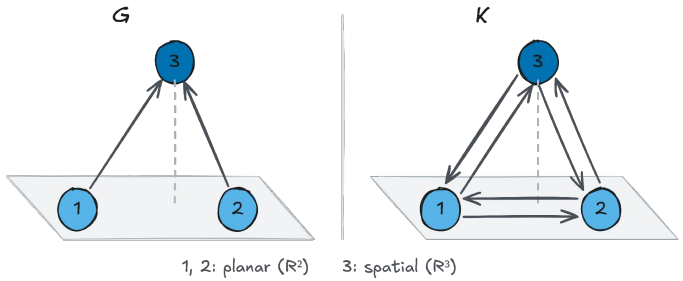
$$\dot{\mathbf{b}}_{\mathcal{G}}(\chi) = \mathbf{B}_{\mathcal{G}}(\chi)\,\delta\chi$$

- Unified framework: all agents in SE(3) (edge-indexed)

$$\mathbf{B}^{\times}_{\mathcal{G}} = \begin{bmatrix} \mathbf{D}_p\,\mathbf{U}\,\bar{\mathbf{E}}^\top & \mathbf{D}_a\,\mathbf{V}\,\bar{\mathbf{E}}_o^\top \end{bmatrix} \in \mathbb{R}^{3m\times 6n}$$

- Node-indexed: DOF restriction per agent

$$\mathbf{B}^{\bullet}_{\mathcal{G}} = \begin{bmatrix} \mathbf{D}_p\,\bar{\mathbf{E}}^\top \bar{S} & \mathbf{D}_a\,\bar{\mathbf{E}}_o^\top \bar{P} \end{bmatrix} \in \mathbb{R}^{3m\times 6n}$$



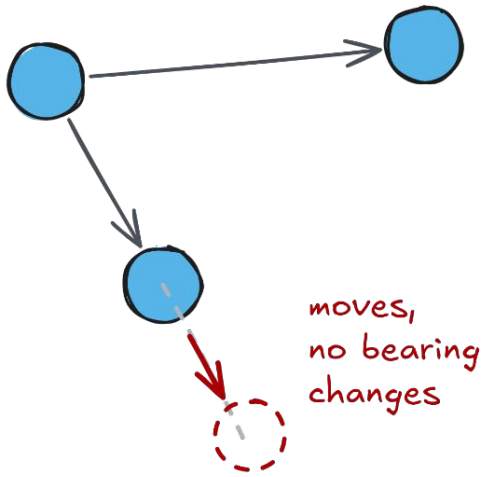
	rk B_G	rk B_K	rank test
edge-idx.	4	5	flexible
node-idx.	4	4	rigid
finite diff.	4	4	rigid

- Edge-indexed: one factor for both endpoints, rank test fails for mixed domains
- Node-indexed: satisfies the requirements, same with edge-indexed on homogeneous formations

[1] S. Zhao, D. Zelazo, IEEE Control Syst. Mag., 2019 [3] G. Michieletto, A. Cenedese, D. Zelazo, IEEE TCNS, 2021 [4] M. H. Trinh, Q. Van Tran, H.-S. Ahn, IEEE TAC, 2020 [28] B. Pozzan et al., IEEE CDC, 2021

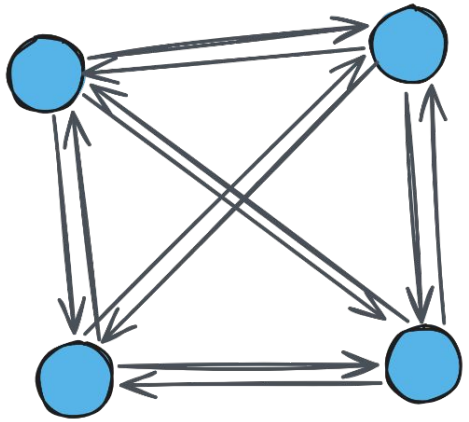
Problem and Approach

too few



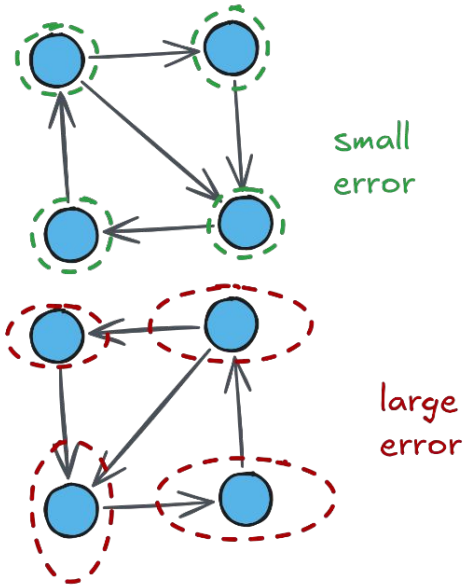
not rigid

all measured



rigid, high measurement cost

same edge count



rigid, different shape
estimation accuracy
under measurement noise

- Competing objectives: sparsity and estimation robustness, subject to rigidity

$$\min_{\mathcal{E}\subseteq(\mathcal{V}\times\mathcal{V})\setminus\Delta} \big(|\mathcal{E}|,\ J_{\text{est}}(\mathcal{G},\chi) \big) \quad \text{subject to} \quad \operatorname{rk}\mathbf{B}_{\mathcal{G}}(\chi) = \operatorname{rk}\mathbf{B}_{\mathcal{K}}(\chi)$$

- Approach: a method modifying the topology sequentially

Structure of the Objectives

Edge count

$$\min_{S\subseteq\mathcal{E}} |S| \quad \text{s.t.} \quad \operatorname{rk}\mathbf{B}_S = \operatorname{rk}\mathbf{B}_{\mathcal{K}}$$

$$f(S) = \operatorname{rk}\mathbf{B}_S = \dim\left(\sum_{e\in S} V_e\right)$$

- Monotone, submodular:

$$f(S\cup\{e\}) - f(S) \;\geq\; f(T\cup\{e\}) - f(T)$$
$$S\subseteq T\subseteq\mathcal{E},\quad e\in\mathcal{E}\setminus T$$

- Minimal rigidity: minimum submodular cover

$$S^\star \in \arg\min_{S\subseteq\mathcal{E}} \left\{|S| : f(S) = f(\mathcal{E})\right\}$$

- (Constructive) Greedy:

$$\frac{|S_{\text{greedy}}|}{|S^\star|} \;\leq\; H(c_{\max}) = \sum_{k=1}^{c_{\max}} \frac{1}{k}$$

- c_{max} = 1 (all planar): exact. c_{max} = 2: within 1.5×.

Rigidity eigenvalue λ

$$\max_{S\subseteq\mathcal{E}} \lambda(S) \quad \text{s.t.} \quad |S| \leq m$$

$$\lambda(S) = \mu_{q+1}\left(\mathbf{B}_S^\top\mathbf{B}_S\right), \qquad q = 6n - \rho_{\mathcal{K}}$$

- Zero until rigid, monotone:

$$\mathbf{B}_{S\cup\{e\}}^\top\mathbf{B}_{S\cup\{e\}} = \mathbf{B}_S^\top\mathbf{B}_S + b_e^\top b_e \;\succeq\; \mathbf{B}_S^\top\mathbf{B}_S$$

- Not submodular: if

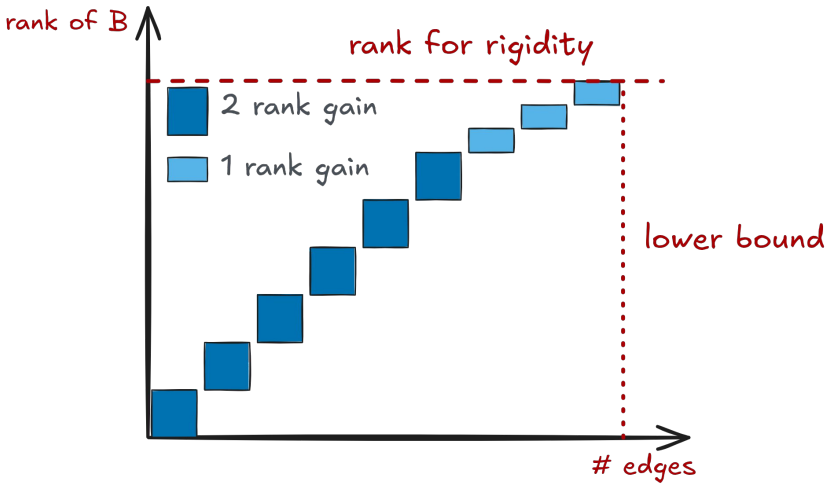
$$\mathcal{G}_{S\cup\{e\}} \text{ flexible}, \quad \mathcal{G}_T \text{ rigid}, \quad b_e v \neq 0$$

$$\lambda(S\cup\{e\}) - \lambda(S) \;=\; 0 \;<\; \lambda(T\cup\{e\}) - \lambda(T)$$

- An edge may contribute nothing before rigidity, but it changes after becoming rigid

Analysis and Tools

Lower bound on the edge count



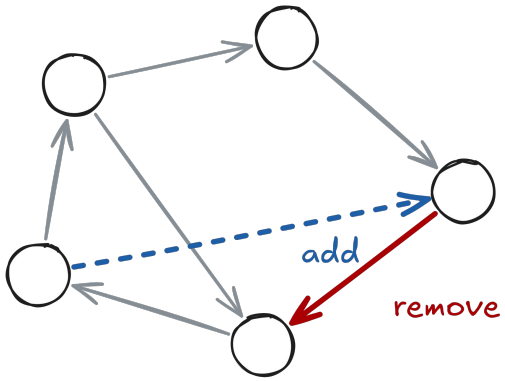
$$m_{\text{req}} = \min\left\{ m \;:\; \sum_{k=1}^{m} c_{(k)} \;\geq\; \rho_{\mathcal{K}} \right\}$$

- Closed form in homogeneous $\mathbb{R}^d$

$$f(n,d) = 1 + kd + r + \mathbf{1}[r > 0]$$

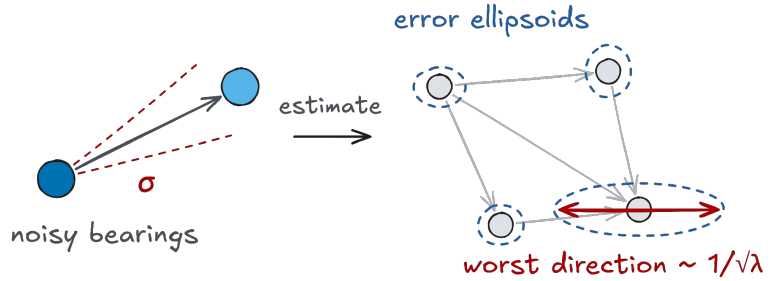
$$k = \left\lfloor \tfrac{n-2}{d-1} \right\rfloor, \quad r = (n-2) \bmod (d-1)$$

Derived quantities (observation)



Add $i \to j$	
$g_{ij} = \text{rk}(b_{ij} Z)$	Rank gain
$\zeta_{ij} = \lVert b_{ij} v \rVert$	Stiffness gain
Remove $i \to j$	
ℓ_{ij}	Rank loss
$\xi_{ij} = 1 - \lambda(\mathbf{B}_{\mathcal{E}\setminus ij})/\lambda$	Conditioning share
$c_{ij} = \text{rk}\, b_{ij} \leq c_{\max}$	Block rank
$f_i,\ \nu_i$	Node freedom, node slack

Estimation error



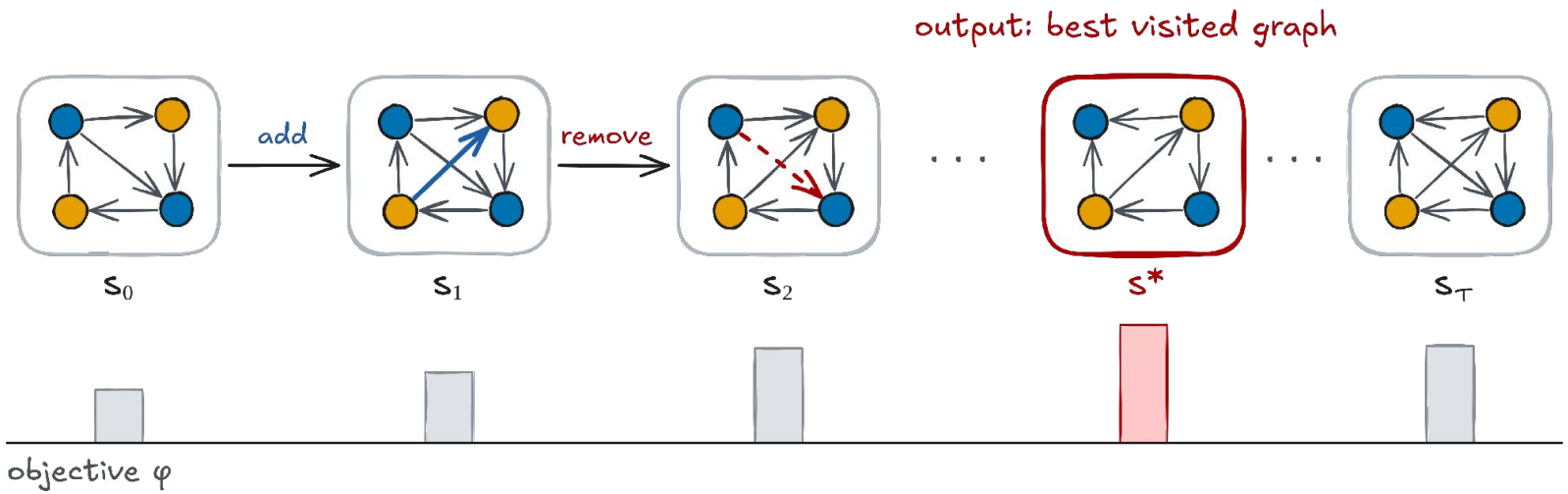
$$\mathcal{I} = \frac{\mathbf{B}^\top \mathbf{B}}{\sigma^2} \qquad \text{Cov}\left(\widehat{\Pi\, \delta\chi}\right) \;\succeq\; \sigma^2 (\mathbf{B}^\top \mathbf{B})^+$$

A	$a_{\text{opt}} = \text{tr}\left((\mathbf{B}^\top \mathbf{B})^+\right) = \sum_k \frac{1}{w_k}$	total MSE
E	$e_{\text{opt}} = 1/\lambda$	worst direction
D	$d_{\text{opt}} = -\sum_k \log w_k$	ellipsoid volume

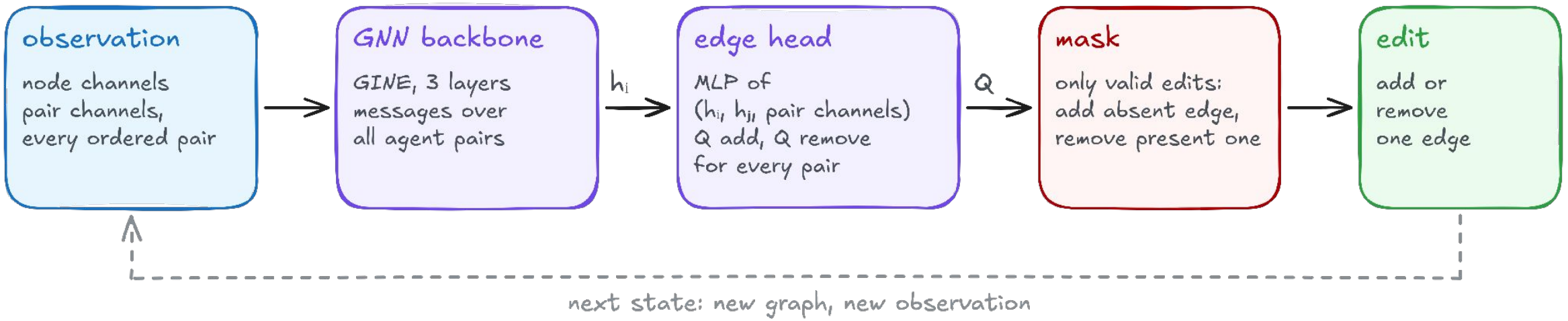
$$\varepsilon(\mathcal{G}, \chi) = \sqrt{\frac{a_{\text{opt}}(\mathbf{B}_s)}{n}}, \qquad a_{\text{opt}} = \text{tr}\left((\mathbf{B}_s^\top \mathbf{B}_s)^+\right)$$

- ε: RMS error per radian of noise
- Comparable across n and scale
- Robustness term of the objective

Graph RL Approach



- State: edge set (poses and domains fixed)
- Action: add or remove one edge
- Reward: change in the objective
- Fixed budget, best visited graph
- Trained with randomized networks, fixed n



Objective and Reward

Objective

$$\varphi(s) = \underbrace{\frac{w_{\text{rank}}\ \text{rk}\,\mathbf{B}}{\rho_{\mathcal{K}}}}_{\text{rigidity}} - \underbrace{\frac{w_{\text{edge}}\, m\, c_{\max}}{\rho_{\mathcal{K}}}}_{\text{sparsity}} + \underbrace{\kappa\, \frac{w_{\text{edge}} c_{\max}}{\rho_{\mathcal{K}}}\, \mathbf{1}[\text{IBR}]\, q(s)}_{\text{robustness}}$$

- Fractions ρ_K: normalization for any n and domain mix
- $(w_{\text{rank}}, w_{\text{edge}}) = (100, 25)$
- Adding an edge (that increases rank) is worth 3x pruning an edge

$$\Delta\varphi\big|_{\Delta r = c_{\max}} = \frac{(w_{\text{rank}} - w_{\text{edge}})\, c_{\max}}{\rho_{\mathcal{K}}} > 0$$

- Rigid graphs, k more edges, scores $q_1 \rightarrow q_2$:

$$\varphi(s_2) - \varphi(s_1) = \frac{w_{\text{edge}} c_{\max}}{\rho_{\mathcal{K}}} \big(\kappa\,(q_2 - q_1) - k\big)$$

- $\kappa \leq 1$: never more edges. $\kappa > 1$: up to κ edges traded for accuracy
- Reward: change in φ, potential-based, dense

$$r_t = \gamma\, \varphi(s_{t+1}) - \varphi(s_t)$$

Robustness score q

Spectral form $q = \text{sigmoid}\left((g - g_{\text{ref}})/\varsigma_g\right)$

g_{ref}: median g of k constructive graphs, at every reset

E-optimality	$g = \log_{10}\lambda$	weakest direction
A-optimality	$g = -\log_{10} a_{\text{opt}}$	total MSE
D-optimality	$g = -d_{\text{opt}}$	error ellipsoid volume

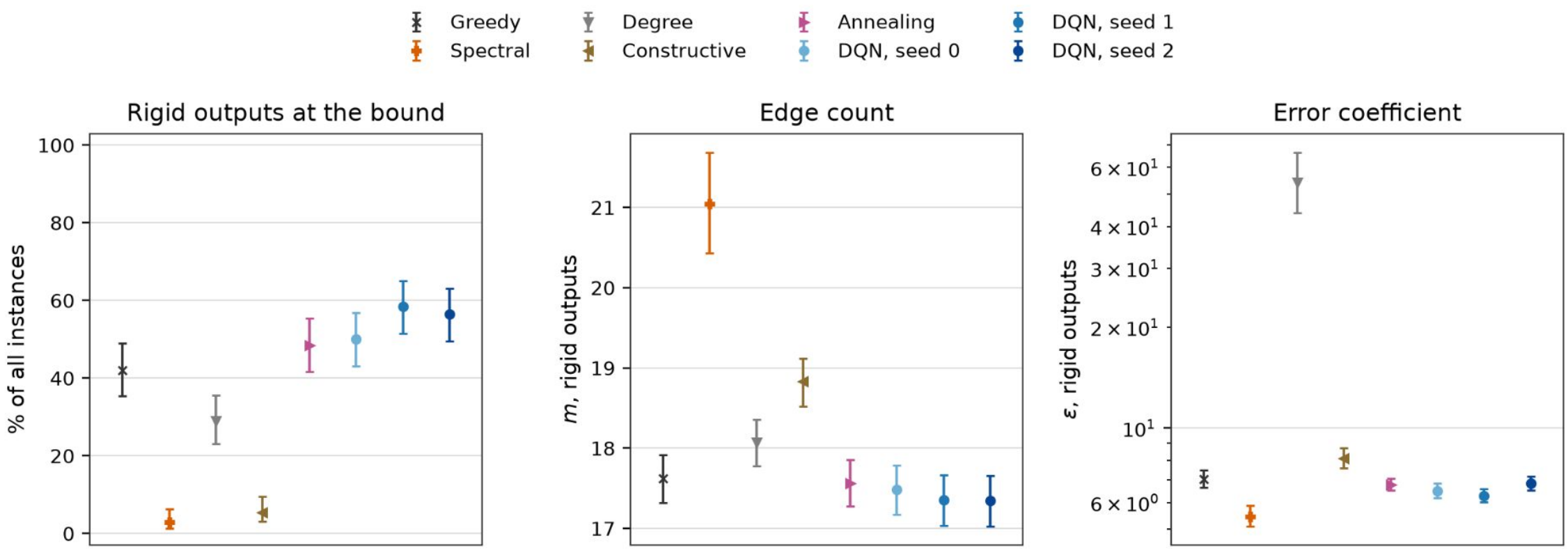
Shape-error form $q = \text{sigmoid}\left(-(\log_{10}\varepsilon - c_\varepsilon)/\varsigma_\varepsilon\right)$

$c_\varepsilon = 1.35$, $\varsigma_\varepsilon = 0.70$: fixed in advance, nothing computed at reset

Shape error	$\varepsilon = \sqrt{a_{\text{opt}}(\mathbf{B}_s)/n}$	RMS error per radian of noise

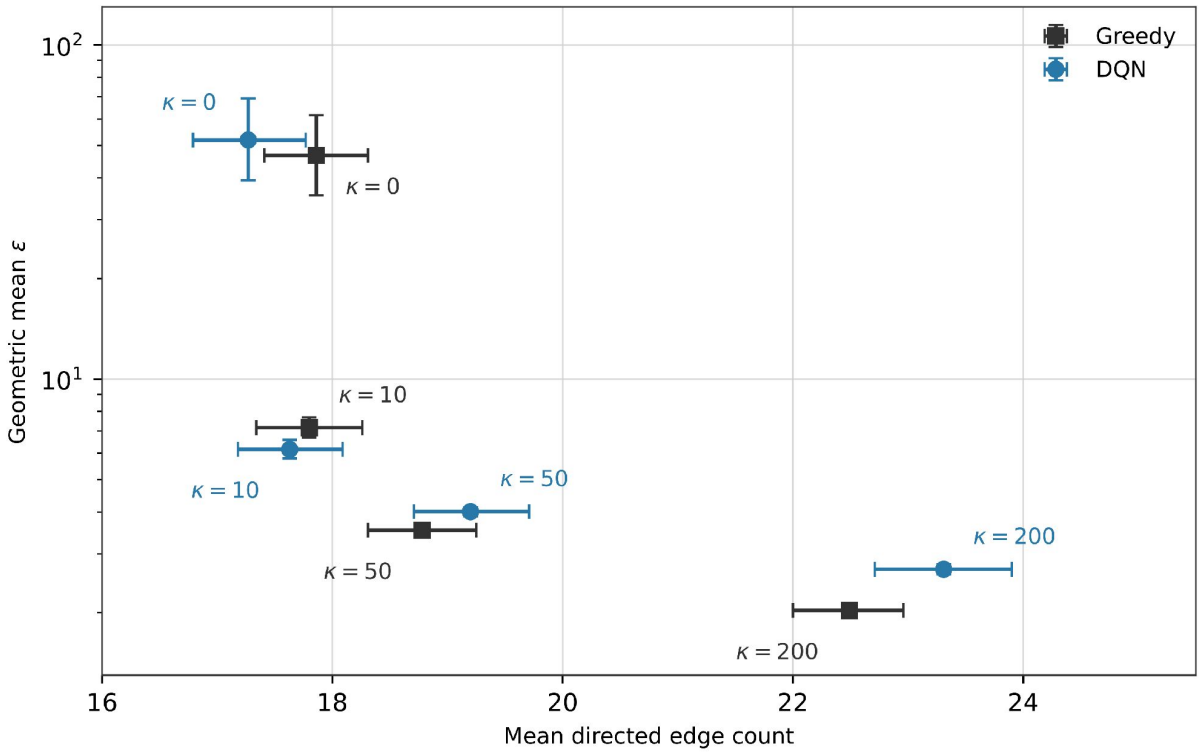
- Only on rigid frameworks
- Trained policies use the shape error form
- Spectral form: reference values would require expensive computation at every reset, cost grows with n

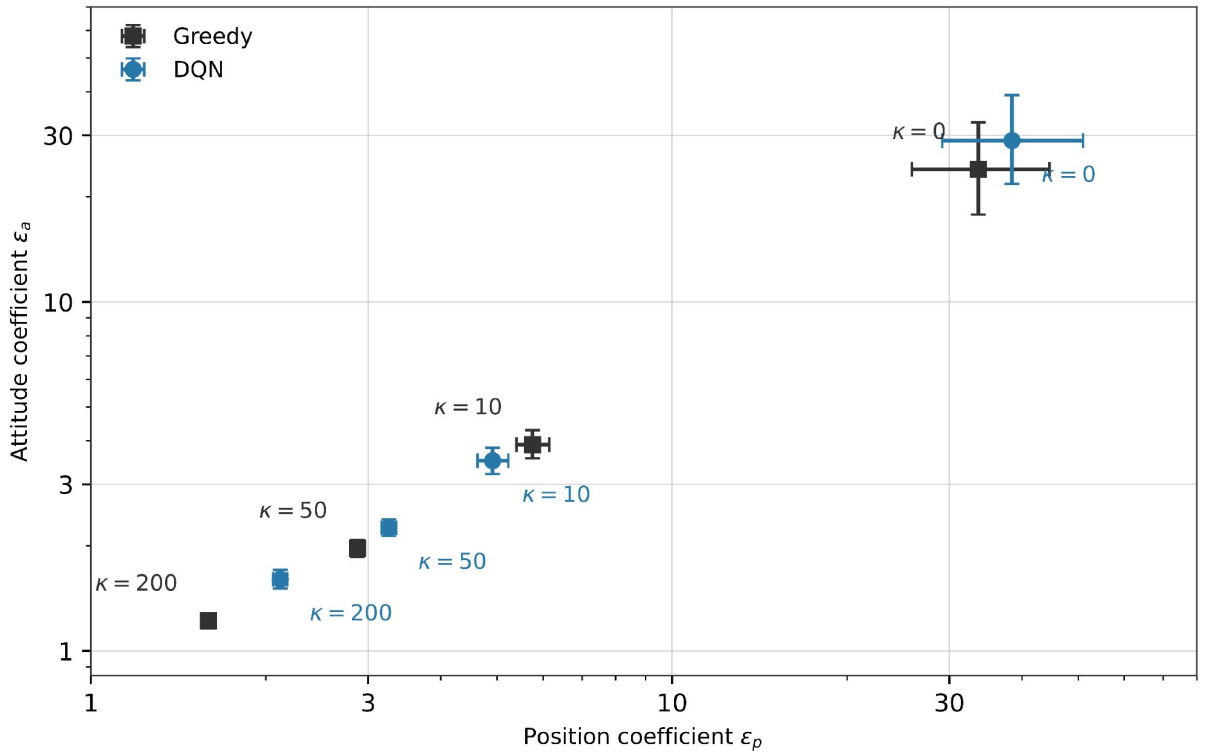
Comparison With Classical Baselines



200 shared $n = 10$ formations with random domains, $\kappa = 10$. Bound attainment over all instances with Wilson intervals, edge count and geometric-mean error over rigid outputs with bootstrap intervals.

Sparsity and Accuracy Trade-off

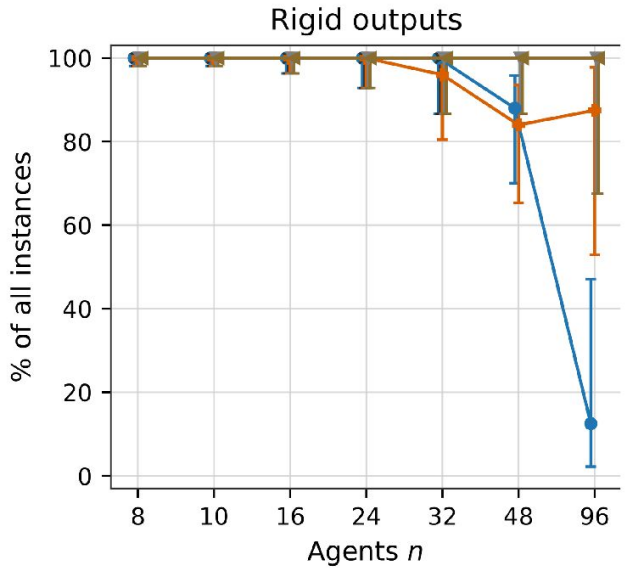


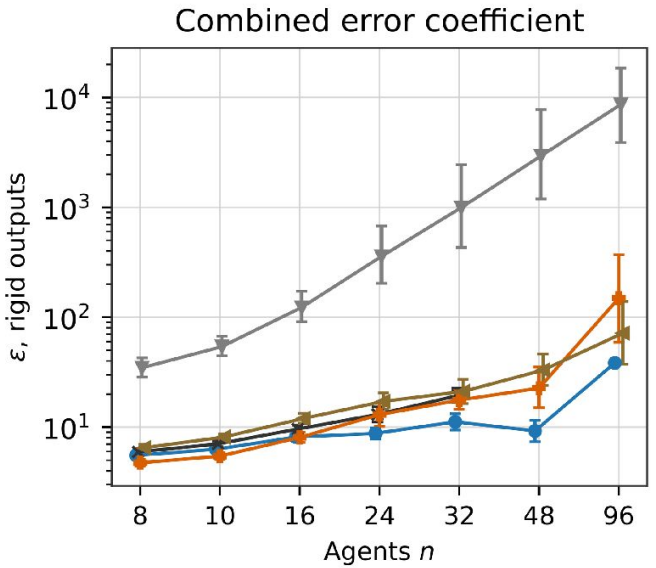


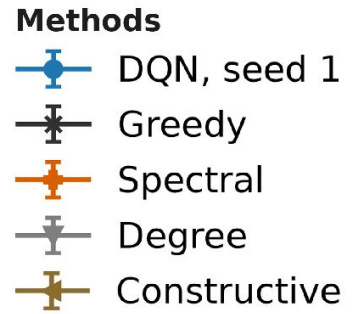
n = 10, 100 formations, each policy and greedy on the objective the policy was trained on

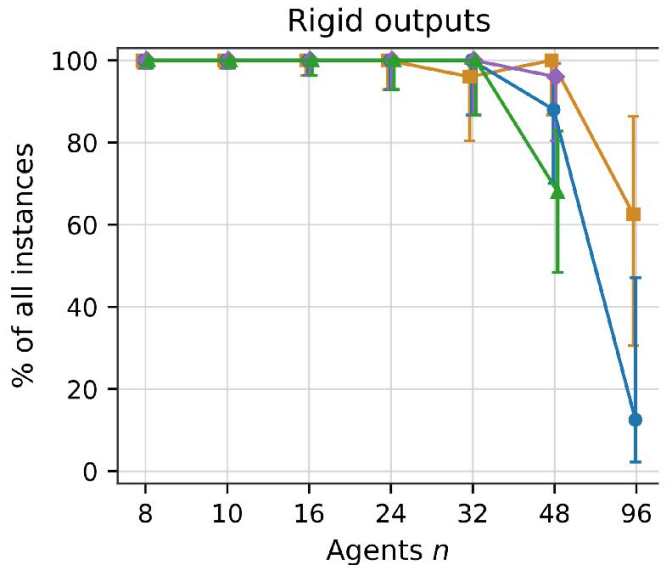
Generalization Across Network Sizes

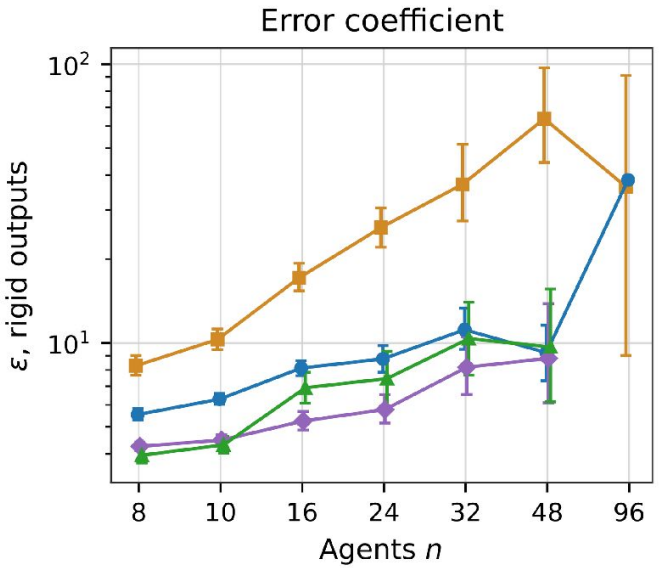
Trained at n = 10, evaluated without retraining

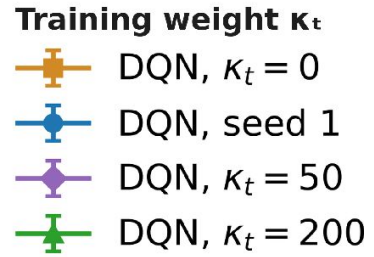




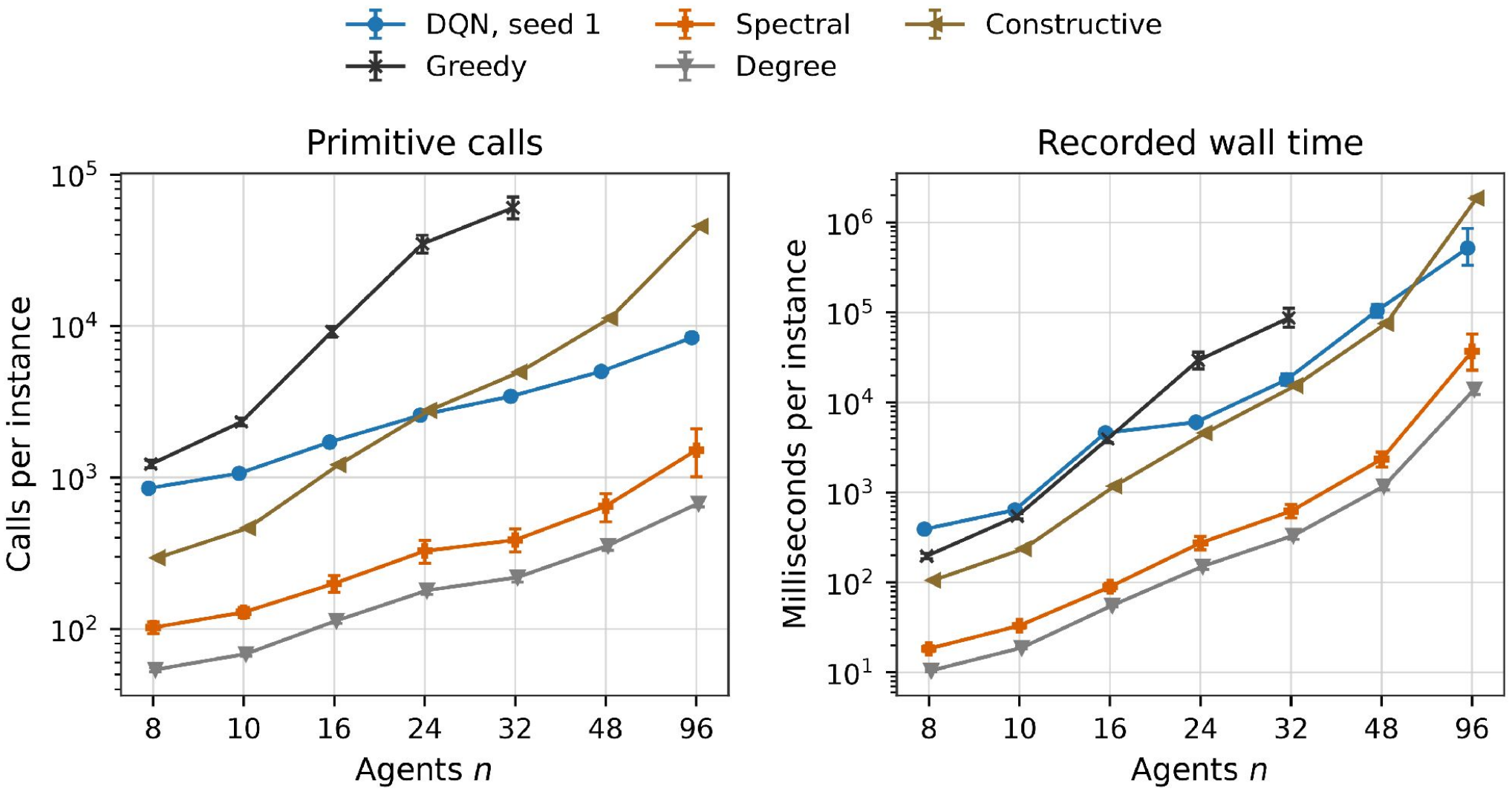








Computational Cost



Per formation

Conclusions

Contributions

- Node-wise extended rigidity matrix for heterogeneous frameworks
- Edge-count bound algorithm, single-edit quantities
- Combinatorial opt. analysis of two objectives/problems
- Topology optimization as sequential process for graph reinforcement learning

Results/Findings

- Marginal improvements at $\kappa = 10$ and $\kappa = 0$, performs worse on other parameter values
- Domain transfer: comparable to greedy
- Size transfer: rigidity fails on the largest network sizes, transfer successful partially
- Policy acts as a fast local search, relies on rigidity-derived inputs
- Cheaper than greedy in terms of algebraic computations

Limitations

- Centralized, global quantities
- Rigidity rewarded, not enforced
- Static formations, exact bearings
- Limited experimental scope

Future work

- Distributed method, subframeworks, CTDE
- Different architectures, hyperparameters, further experimentation
- Longer horizon decision making with rigidity guarantees



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Thank you

Noyan Erdin Kilic

Backup: Heterogeneous Rigidity Matrix

- Unified framework: all agents in SE(3)

$$\mathbf{B}_{\mathcal{G}}^{\times} = \begin{bmatrix} \mathbf{D}_p \mathbf{U} \bar{\mathbf{E}}^{\top} & \mathbf{D}_a \mathbf{V} \bar{\mathbf{E}}_o^{\top} \end{bmatrix} \in \mathbb{R}^{3m \times 6n}$$

edge-indexed:

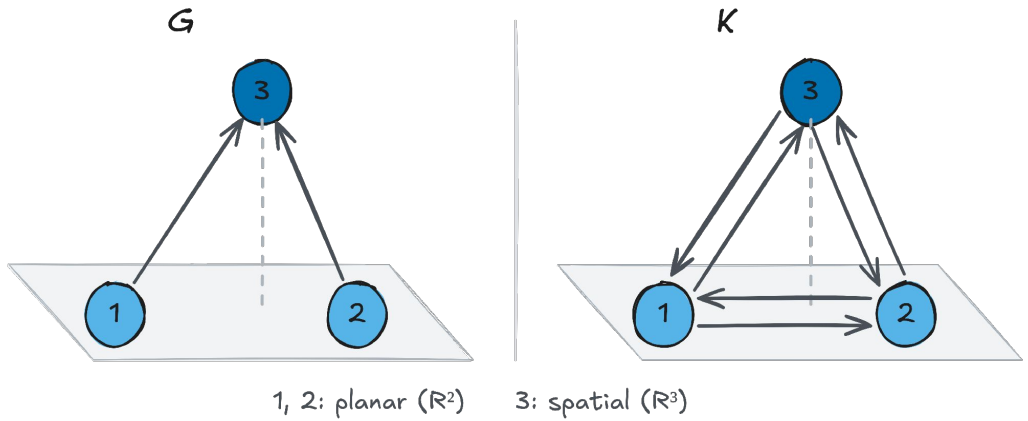
$$\left[\mathbf{D}_p \, \mathbf{U} \, \bar{\mathbf{E}}^{\top}\right]_{k,\cdot} = \left[\, \mathbf{0} \cdots \underbrace{-\mathbf{D}_p^{(k)} \mathbf{U}_{ij}}_{i} \cdots \underbrace{+\mathbf{D}_p^{(k)} \mathbf{U}_{ij}}_{j} \cdots \mathbf{0} \,\right]$$

- Node-indexed: DOF restriction per node

$$\boxed{\mathbf{B}_{\mathcal{G}}^{\bullet} = \begin{bmatrix} \mathbf{D}_p \bar{\mathbf{E}}^{\top} \bar{S} & \mathbf{D}_a \bar{\mathbf{E}}_o^{\top} \bar{P} \end{bmatrix} \in \mathbb{R}^{3m \times 6n}}$$

node-indexed:

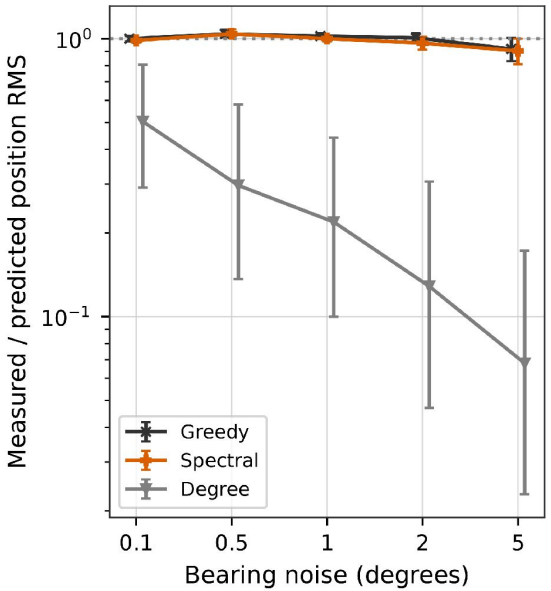
$$\left[\mathbf{D}_p \bar{\mathbf{E}}^{\top} \bar{S}\right]_{k,\cdot} = \left[\, \mathbf{0} \cdots \underbrace{-\mathbf{D}_p^{(k)} S_i}_{i} \cdots \underbrace{+\mathbf{D}_p^{(k)} S_j}_{j} \cdots \mathbf{0} \,\right]$$

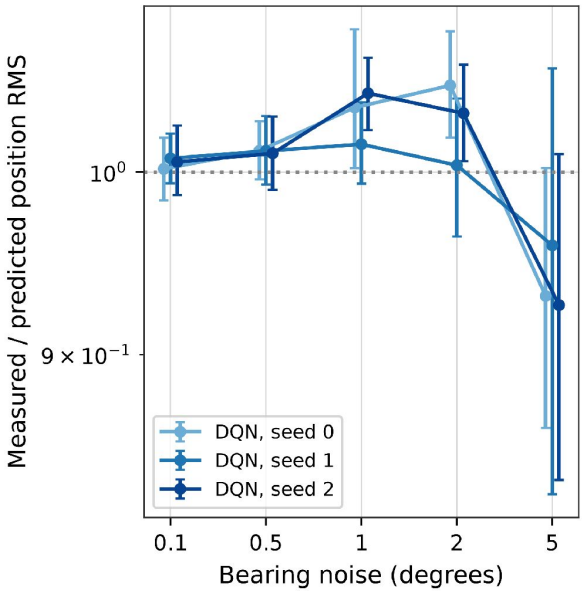


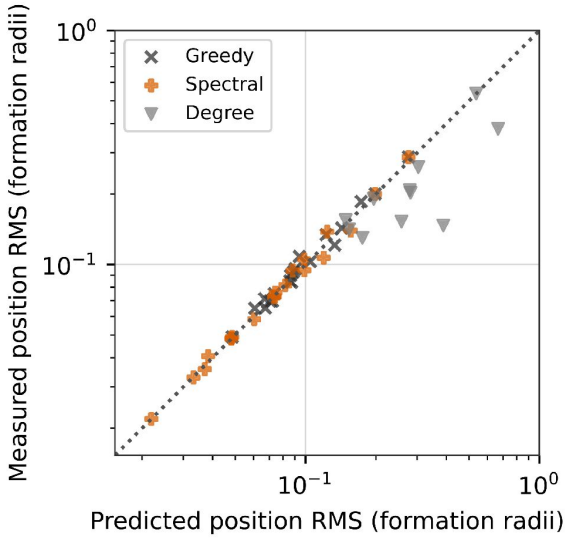
	rk B_G	rk B_K	rank test
edge-indexed	4	**5**	**flexible**
node-indexed	4	4	**rigid**
finite differences	4	4	**rigid**

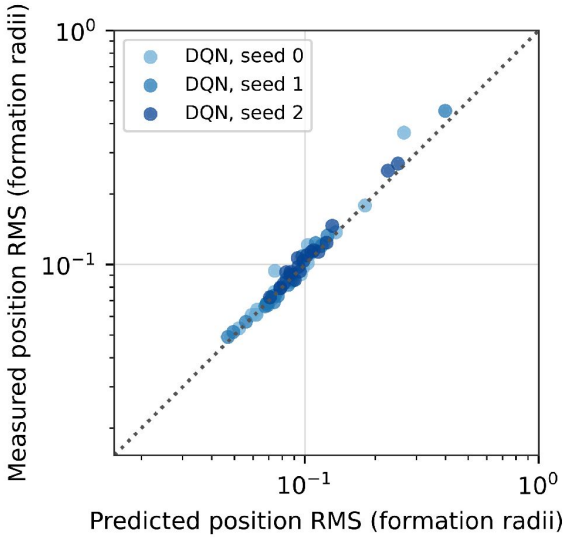
extra rank on K come from a vertical motion of the planar agents

Backup: Error Prediction Under Noise









Measured (Gauss-Newton recovery, 100 noise draws) vs predicted position RMS. 20 formations per method

Linear prediction holds while $\sigma\varepsilon \lesssim 0.1$, overstates the error otherwise

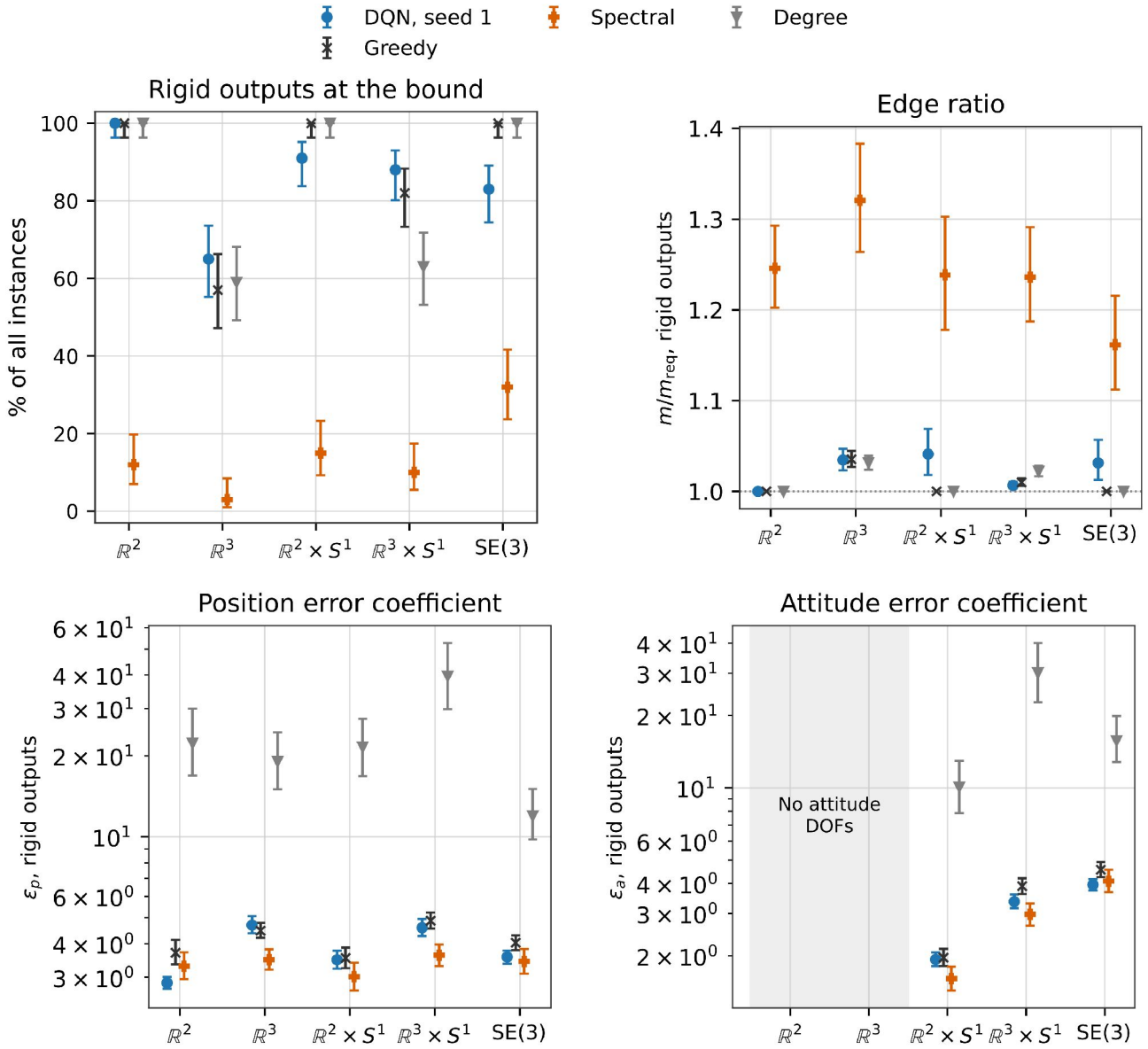
Backup: Baseline Methods

Method	Search step	Stops	Returns
Greedy	Evaluate every single-edge toggle, take the largest improvement	No improvement or edit limit	Final graph
Spectral	Rank edits by rank and spectral quantities, evaluate the best 5	No improvement among the 5, or edit limit	Final graph
Degree	Add edges by endpoint degree until rigid, then remove edges keeping rigidity	No removable edge	Final graph
Constructive	From the empty graph, random candidate order, keep rank-increasing additions	Rigid	Best of 5 runs
Annealing	Single-edge edits, worsening ones accepted w.p. $\exp(-\Delta/T)$	Greedy's evaluation count	Best visited graph
DQN	Policy selects the edit, no candidate evaluation	Horizon T_{max}	Best visited graph

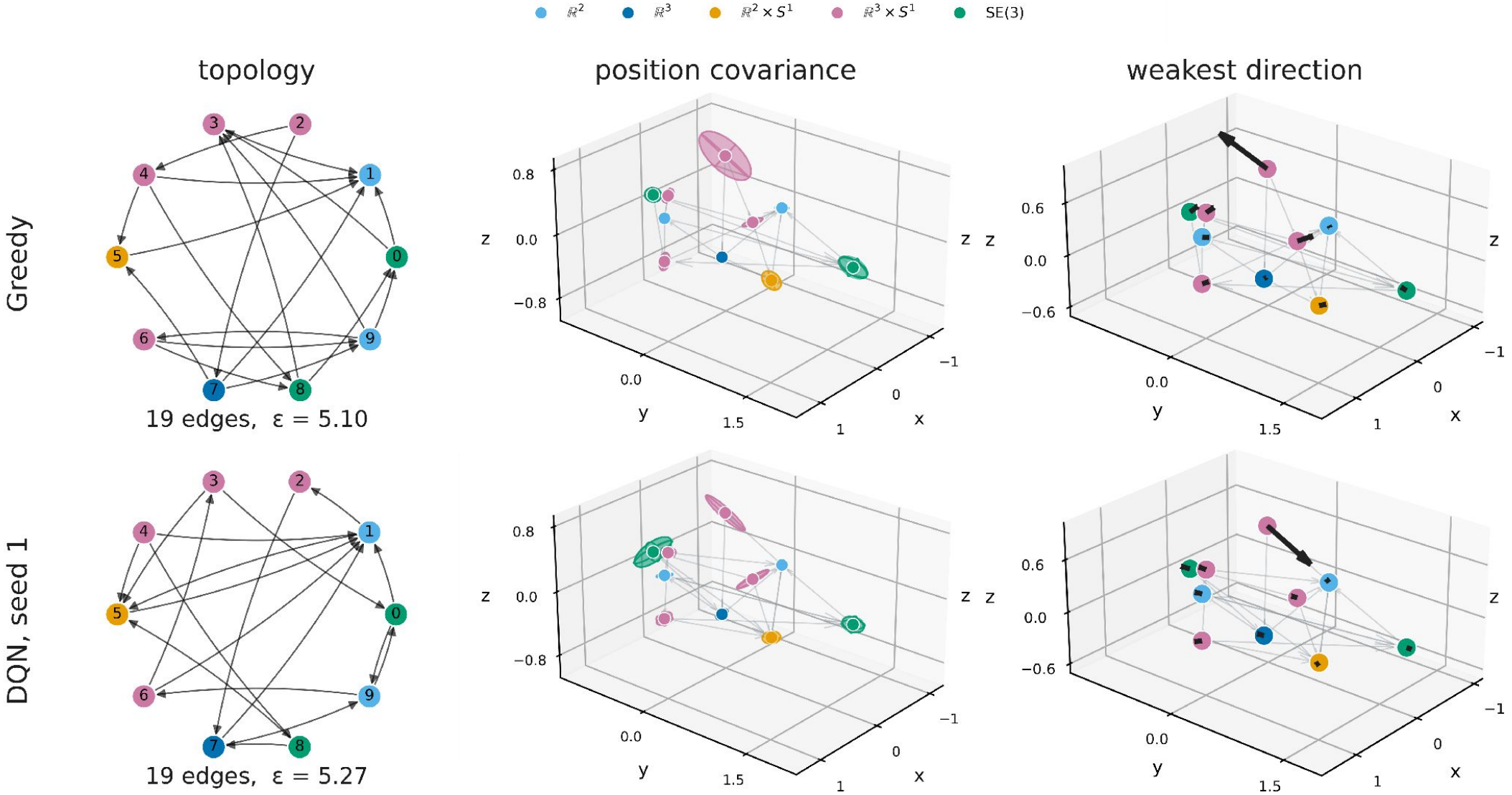
Same poses and domains for all, edit methods share the initial graph (constructive starts from empty graph), outputs scored on the same objective

Backup: Comparison Across Agent Domains

Homogeneous domains, n = 10, 100 formations each, policy trained on random domain mixes



Backup: Resulting Topologies



Ellipsoids: position covariance blocks of the bound, same scale in both rows. Arrows: weakest direction, eigenvector of the smallest positive eigenvalue (scaled per panel).